

LAMPANG, Thailand

WMO: 483280

Lat: 18.2781N

Lon: 99.5064E

Elev: 244

StdP: 98.43

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |             |            |             | Coldest Month WS/MCDB |             |            |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|-------------|-------------|------------|-------------|-----------------------|-------------|------------|-------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |            |             | 99%         |            |             | 0.4%                  |             | 1%         |             |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP          | HR         | MCDB        | WS                    | MCDB        | WS         | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>12     | (b)<br>12.8 | (c)<br>14.3 | (d)<br>8.9                    | (e)<br>7.3 | (f)<br>26.9 | (g)<br>10.5 | (h)<br>8.1 | (i)<br>24.5 | (j)<br>4.0            | (k)<br>26.9 | (l)<br>3.4 | (m)<br>27.5 | (n)<br>0.1            | (o)<br>120 | (p)<br>0.294 |

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 4             | 13.6                   | 39.2            | 23.9 | 38.0 | 23.6 | 36.8 | 23.7 | 27.4                | 32.2 | 26.9 | 32.1 | 26.5 | 31.8 | 1.4                  | 180  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 26.2                            | 22.3 | 29.3 | 25.7 | 21.7 | 28.9 | 25.2 | 20.9 | 28.4 | 88.8          | 32.2 | 86.4 | 32.2 | 84.6 | 32.1 | 34.1              |

Extreme Annual Design Conditions

| Extreme Annual WS |     |      | Extreme Annual Temperature |      |      |                    | n-Year Return Period Values of Extreme Temperature |           |      |            |      |            |      |            |      |  |
|-------------------|-----|------|----------------------------|------|------|--------------------|----------------------------------------------------|-----------|------|------------|------|------------|------|------------|------|--|
| 1%                |     | 2.5% | 5%                         | Mean |      | Standard Deviation |                                                    | n=5 years |      | n=10 years |      | n=20 years |      | n=50 years |      |  |
| Min               |     | Max  |                            | Min  | Max  |                    | Min                                                | Max       | Min  | Max        | Min  | Max        | Min  | Max        |      |  |
| (a)               | (b) | (c)  |                            | (e)  | (f)  | (g)                | (h)                                                | (i)       | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |
| 4.7               | 3.8 | 3.2  | DB                         | 10.0 | 41.0 | 2.0                | 1.1                                                | 8.6       | 41.8 | 7.4        | 42.4 | 6.3        | 43.1 | 4.8        | 43.9 |  |
|                   |     |      | WB                         | 9.6  | 29.1 | 2.0                | 1.7                                                | 8.2       | 30.3 | 7.0        | 31.3 | 5.9        | 32.2 | 4.5        | 33.4 |  |

Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 27.4   | 23.5  | 25.7  | 28.5  | 30.7  | 29.7  | 29.1  | 28.5  | 28.2  | 28.1  | 27.4  | 25.6  | 23.3  |
|                                                                                  | DBStd       | 2.84   | 1.98  | 1.93  | 1.95  | 1.95  | 2.03  | 1.24  | 1.30  | 1.12  | 1.09  | 1.35  | 1.91  | 2.37  |
|                                                                                  | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                                  | HDD18.3     | 2      | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 2     |
|                                                                                  | CDD10.0     | 6334   | 418   | 439   | 575   | 622   | 610   | 573   | 573   | 563   | 542   | 538   | 468   | 413   |
|                                                                                  | CDD18.3     | 3294   | 161   | 206   | 317   | 372   | 352   | 323   | 315   | 304   | 292   | 280   | 218   | 156   |
|                                                                                  | CDH23.3     | 35420  | 1440  | 2245  | 3948  | 4958  | 4196  | 3711  | 3402  | 3071  | 2831  | 2531  | 1829  | 1260  |
| CDH26.7                                                                          | 16566       | 597    | 1172  | 2297  | 2953  | 2131  | 1644  | 1373  | 1141  | 1065  | 991   | 736   | 466   |       |
| Wind                                                                             | WSAvg       | 0.6    | 0.4   | 0.5   | 0.6   | 0.8   | 0.7   | 0.9   | 0.8   | 0.7   | 0.5   | 0.4   | 0.3   | 0.4   |
| Precipitation                                                                    | PrecAvg     | 1132   | 6     | 7     | 23    | 63    | 156   | 127   | 137   | 223   | 224   | 126   | 34    | 7     |
|                                                                                  | PrecMax     | 1520   | 58    | 57    | 57    | 229   | 309   | 230   | 332   | 413   | 328   | 257   | 134   | 28    |
|                                                                                  | PrecMin     | 728    | 0     | 0     | 0     | 1     | 24    | 40    | 56    | 126   | 114   | 49    | 0     | 0     |
|                                                                                  | PrecStd     | 211    | 14    | 14    | 19    | 59    | 70    | 52    | 73    | 69    | 68    | 54    | 35    | 8     |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 34.2  | 37.2  | 39.7  | 41.3  | 40.0  | 36.9  | 35.8  | 34.9  | 35.0  | 34.9  | 34.7  | 33.5  |
|                                                                                  |             | MCWB   | 20.9  | 21.5  | 22.2  | 24.2  | 24.9  | 26.0  | 25.8  | 25.9  | 26.0  | 26.0  | 24.2  | 22.7  |
|                                                                                  | 2%          | DB     | 32.8  | 35.8  | 38.3  | 40.0  | 37.9  | 35.2  | 34.2  | 33.6  | 33.8  | 33.6  | 33.2  | 32.1  |
|                                                                                  |             | MCWB   | 21.0  | 20.9  | 22.1  | 23.9  | 25.5  | 25.8  | 25.7  | 26.0  | 26.2  | 25.7  | 24.0  | 22.4  |
|                                                                                  | 5%          | DB     | 31.6  | 34.6  | 37.2  | 38.7  | 36.2  | 34.0  | 33.2  | 32.6  | 32.8  | 32.6  | 32.1  | 30.9  |
|                                                                                  |             | MCWB   | 21.0  | 21.1  | 22.0  | 24.2  | 25.7  | 25.9  | 25.6  | 25.8  | 25.9  | 25.5  | 24.0  | 22.1  |
|                                                                                  | 10%         | DB     | 30.3  | 33.2  | 35.9  | 37.1  | 34.7  | 32.9  | 32.2  | 31.7  | 31.7  | 31.5  | 30.9  | 29.5  |
|                                                                                  |             | MCWB   | 20.8  | 21.1  | 22.1  | 24.4  | 25.6  | 25.7  | 25.5  | 25.6  | 25.9  | 25.2  | 23.6  | 21.6  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 24.2  | 24.7  | 26.5  | 28.8  | 28.7  | 27.8  | 27.4  | 27.2  | 27.4  | 27.1  | 26.2  | 25.1  |
|                                                                                  |             | MCDB   | 29.9  | 32.4  | 31.9  | 34.7  | 32.6  | 31.7  | 31.6  | 31.6  | 32.1  | 32.1  | 31.0  | 30.6  |
|                                                                                  | 2%          | WB     | 23.2  | 23.7  | 25.0  | 26.7  | 27.1  | 26.9  | 26.6  | 26.7  | 26.8  | 26.5  | 25.5  | 24.1  |
|                                                                                  |             | MCDB   | 28.9  | 31.0  | 32.7  | 33.6  | 33.3  | 32.2  | 31.6  | 31.2  | 31.6  | 31.5  | 30.4  | 29.4  |
|                                                                                  | 5%          | WB     | 22.4  | 23.1  | 24.2  | 25.8  | 26.5  | 26.4  | 26.2  | 26.2  | 26.5  | 26.1  | 25.0  | 23.2  |
|                                                                                  |             | MCDB   | 28.2  | 30.3  | 32.1  | 34.1  | 32.8  | 31.7  | 31.2  | 30.7  | 31.0  | 30.9  | 29.8  | 28.2  |
|                                                                                  | 10%         | WB     | 21.7  | 22.3  | 23.7  | 25.3  | 26.0  | 26.1  | 25.8  | 25.9  | 26.1  | 25.7  | 24.4  | 22.5  |
|                                                                                  |             | MCDB   | 27.6  | 29.5  | 31.6  | 33.4  | 32.4  | 31.3  | 30.6  | 30.3  | 30.4  | 30.1  | 28.9  | 27.3  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 15.1  | 16.7  | 15.7  | 13.6  | 10.0  | 8.5   | 7.8   | 7.7   | 8.2   | 9.2   | 11.4  | 13.4  |
|                                                                                  |             | MCDBR  | 16.8  | 18.4  | 17.8  | 15.5  | 12.5  | 10.2  | 9.5   | 9.2   | 9.8   | 10.7  | 12.8  | 14.5  |
|                                                                                  |             | MCWBR  | 6.1   | 5.7   | 4.9   | 3.9   | 3.5   | 3.0   | 3.0   | 3.1   | 3.2   | 3.7   | 4.6   | 5.5   |
|                                                                                  | 5% WB       | MCDBR  | 13.3  | 14.9  | 14.4  | 13.0  | 10.4  | 8.9   | 8.4   | 8.2   | 8.8   | 9.5   | 10.7  | 12.3  |
| MCWBR                                                                            |             | 5.2    | 5.3   | 4.9   | 4.5   | 3.7   | 3.2   | 3.1   | 3.1   | 3.3   | 3.6   | 4.2   | 5.0   |       |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |        | 0.448 | 0.532 | 0.667 | 0.653 | 0.501 | 0.442 | 0.431 | 0.445 | 0.480 | 0.527 | 0.447 | 0.443 |
|                                                                                  | taud        |        | 2.173 | 1.953 | 1.657 | 1.746 | 2.183 | 2.379 | 2.407 | 2.348 | 2.233 | 2.058 | 2.253 | 2.212 |
|                                                                                  | Ebn at Noon |        | 828   | 775   | 687   | 703   | 808   | 850   | 860   | 855   | 825   | 770   | 824   | 819   |
|                                                                                  | Edh at Noon |        | 141   | 184   | 256   | 236   | 150   | 122   | 119   | 127   | 142   | 164   | 129   | 132   |
| All-Sky Solar<br>Radiation                                                       | RadAvg      |        | 4.56  | 5.27  | 5.58  | 6.04  | 5.50  | 4.89  | 4.57  | 4.50  | 4.65  | 4.52  | 4.62  | 4.33  |
|                                                                                  | RadStd      |        | 0.23  | 0.21  | 0.35  | 0.33  | 0.47  | 0.37  | 0.31  | 0.28  | 0.21  | 0.25  | 0.30  | 0.23  |

Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | +1      | N/A     | N/A     |
| Regional (21 neighbors) | +0.26 | N/A     | N/A    | N/A     | N/A   | +0.15 | N/A         | N/A     | +121    | +121    |

Nomenclature: See separate page